

RAM KUMAR TAMANG

Python Backend Engineer · Django & DRF Specialist · Geospatial & Disaster Risk Systems

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PROFESSIONAL SUMMARY

Python Backend Engineer with 3+ years of professional experience building scalable systems across geospatial, disaster risk, and telecom domains. At NAXA Pvt. Ltd., delivering national-scale weather forecast and flood early-warning platforms used by the Department of Hydrology & Meteorology, local governments, and humanitarian organisations across Nepal, Bangladesh, and Malawi. Previously at KrispCall (YC-backed), built VoIP microservices serving 10,000+ concurrent users, integrated 6+ CRM platforms, and managed Stripe payments and async event-driven pipelines. Core stack: Django, FastAPI, PostgreSQL/PostGIS, Redis, Celery, Docker.

TECHNICAL SKILLS

Languages & Frameworks: Python (primary), Django, DRF, FastAPI, Starlette, Flask, JavaScript

Databases, Caching & Task Queues: PostgreSQL/PostGIS, MongoDB, Elasticsearch, Redis, SQLite, Celery

APIs & Integrations: REST, GraphQL, gRPC, Stripe, Khalti, eSewa, Webhook handling, CRM platforms (Pipedrive, Intercom, MS Dynamics)

Geospatial & Data: GeoNode, GeoServer, ECMWF, NOAA GFS, ICON-EPS, GLOFAS, GeoGLOWS, Pandas, NumPy, Plotly

DevOps & Cloud: Docker, GitHub Actions (CI/CD), AWS EC2, GCP, Nginx, SonarQube

Auth & Security: OAuth2, JWT, rate limiting, API throttling, data encryption

AI & Robotics: OpenAI GPT/Whisper, Google Gemini/STT, Kokoro TTS, DeepFace, LangChain, multi-service orchestration

Monitoring & Practices: Grafana, Datadog · Domain-Driven Design, SOLID, TDD (PyTest), Agile/Scrum (Jira)

PROFESSIONAL EXPERIENCE

Backend Engineer (Python / Geospatial) · NAXA Pvt. Ltd.

Jan 2024 – Present

Kathmandu, Nepal · Onsite

- Led backend development of Nepal's national weather forecast system for the Department of Hydrology & Meteorology (DHM), integrating multi-source data from ECMWF, NOAA GFS, ICON-EPS, HIWAT, GLOFAS, and GeoGLOWS.
- Own and maintain the National Geoportal of Nepal (nationalgeoportal.gov.np), a GeoNode-based national geospatial platform; led its migration from GeoNode 3.x → 4.4.5, including database migration, GeoServer/GeoFence security-rule synchronisation, and production cutover.
- Built the backend weather module and real-time alerting system for DASTAA — a flood anticipatory action platform operational across Nepal, Bangladesh, and Malawi, used by local governments and humanitarian organisations.
- Integrated SMS mass-alerting (Aakash Tel) and IVR voice notifications (Malai Phase Tel), enabling household-level early-warning delivery before flood events.
- Implemented bias correction and trend analysis pipelines using linear regression and interpolation to improve forecast accuracy across Nepal's river monitoring network.
- Built async background job processing with Celery and Redis for scheduled forecast ingestion, data transformation, and batch report generation.
- Developed high-performance REST APIs with FastAPI and Django; managed PostgreSQL/PostGIS with indexing and query optimisation for large geospatial datasets.
- Containerised all services with Docker; used Elasticsearch and MongoDB for log ingestion and unstructured geospatial data.

Associate Software Engineer · KrispCall (YC-backed)

Aug 2023 – Dec 2023

Kathmandu, Nepal · Onsite

- Designed and maintained VoIP microservices in Python/Django ensuring 99.9% uptime for 10,000+ concurrent users, applying Domain-Driven Design and SOLID principles.
- Integrated 6+ CRM platforms (Pipedrive, Intercom, Microsoft Dynamics) with webhook-based patterns including validation, idempotency, and retry mechanisms — improving client workflows by 25%.
- Built RESTful and GraphQL APIs for telephony features; implemented event-driven architecture with Python Signals and gRPC for high-throughput async processing.
- Integrated Stripe payment APIs and managed full transaction lifecycle including webhooks, retries, and failure handling.

- Deployed on Amazon EC2; automated CI/CD with GitHub Actions; containerised workloads with Docker; enforced code quality via SonarQube; monitored with Grafana and Datadog.
- Mentored junior developers; authored comprehensive API and system documentation that significantly reduced team onboarding time.

Django Developer · Robotics Association of Nepal

Nov 2021 – Nov 2022

Lalitpur, Nepal · Onsite

- Built Django web applications for robotics data tracking and real-time analytics serving 50+ researchers.
- Processed 1M+ sensor data points via ML-ready pipelines; developed interactive Plotly dashboards for data visualisation.

KEY PROJECTS

National Geoportal of Nepal · GeoNode · GeoServer · Django · PostgreSQL/PostGIS · Nginx

Nepal's national geospatial data-sharing platform (nationalgeoportal.gov.np). Led the GeoNode 3.x → 4.4.5 migration and production cutover; resolved GeoFence security gaps, GeoServer/nginx routing misconfigurations, and authentication issues; built preflight and schema-diff tooling to make future migrations repeatable and safe.

Nepal Weather Forecast & River Monitoring — DHM · FastAPI · Django · Celery · PostgreSQL/PostGIS

National-level multi-source forecast platform integrating ECMWF, NOAA GFS, ICON-EPS, GLOFAS, GeoGLOWS, HIWAT, and DHM data. Implemented bias correction (linear regression, interpolation) and trend analysis; built scheduled ingestion jobs with real-time alerting via API and SMS.

DASTAA — Digital & Spatial Technologies for Anticipatory Action · Django · FastAPI · Redis · Docker · Elasticsearch

Flood anticipatory action platform (Nepal, Bangladesh, Malawi) for NAXA & IHRR. Built the weather data module, household-level risk profiling backend, and real-time SMS/IVR alert dispatch integrations (Aakash Tel, Malai Phase Tel).

InMoov Robot — Software Lead · Python · OpenAI Whisper · VLM · LLMs · Kokoro TTS · DeepFace

Open-source upper-body robotic system at HTP Lab, Kathmandu. Built a Python master control system integrating dual-camera vision with VLM-powered environment understanding, OpenAI Whisper STT, LLM reasoning (OpenAI GPT primary, Google Gemini fallback), Kokoro TTS voice output, DeepFace for face and emotion recognition, magnetic encoders for joint position sensing, and accelerometer-based orientation awareness — enabling the robot to speak, understand its environment, and execute left/right movement with full positional awareness.

OzNepal E-Commerce Platform · Django · DRF · Stripe · PostgreSQL

Full cart, checkout, and payments backend with Stripe integration — payment webhooks, transaction lifecycle management, and retry logic for failed payments.

Biomedical Equipment Management System (BEMS) · Django · SQLite · React · Celery

Reduced hospital equipment downtime across 100+ hospitals. Implemented batch scheduling and real-time notifications for equipment allocation and maintenance.

Humanizer — AI Writing Assistant · Django · Khalti · eSewa · AI APIs

Web platform providing AI-assisted text improvement for Nepali students. Integrated Khalti and eSewa payment gateways with full checkout-to-access provisioning lifecycle.

EDUCATION

Data Science (Applied)

2025 – Present

WorldQuant University

Bachelor of Information & Communication Technology

2019 – 2023

Asia e University

CERTIFICATIONS & ACHIEVEMENTS

- Large Language Models Certification (Ongoing) — Udemy, 2025
- Microsoft Certifications — Dynamics 365, Microsoft Teams, Innovative Educator
- Hackathon Winner — CodeBrisk, SecurityPal, Udhyam Innovation Challenge
- Finalist — ICT Rising Star Award, ADB Hackathon
- Hackathon Judge & AI Trainer — Jhapa Dev Program, 2026; led AI/ML bootcamp for 100+ school and college students as sponsored trainer; connected with HISSAN Nepal leadership and college administrators
- Delivered AI and technology training across students, professionals, and institutions — including professional AI upskilling for the Mission Oxygen COVID-19 response team, NIDN, and IME Group; robotics workshops at multiple colleges; and 10+ Python/web development sessions reaching 1,000+ participants.